

Analysis of Algorithms

Assignment 02

1. Solve the following recurrences. The solution of each exercise includes finding the closed form and then verifying using mathematical induction. The solution of the recurrences can be done by hand and then scanned to a PDF file.

1.1. $T(n) = 3 T(n/2) + n$, $T(1) = 0$, for all n powers of two, 1, 2, 4, 8, 16...

1.2. $T(n) = 3 T(n-1) + 4 T(n-2)$, $T(0) = 0$, $T(1) = 5$

1.3. $T(n) = 5 T(n-1) - 6 T(n-2)$, $T(0) = 0$, $T(1) = 1$

1.4. $x(n) = 2 x(n-1) + 4 x(n-2)$, $x(0) = 1$, $x(1) = 2$, for $n > 1$

2. The Fibonacci sequence is generated using the following recurrence:

$$T(n) = T(n-2) + T(n-1)$$

2.1. Write a C++ program that generates the sequence using iteration.

2.2. Write a C++ program that generates the sequence using recursion.

Submission

All submissions are done via the Moodle platform. Make sure to upload within the submission period.